Figure 3 supplementary methods: linear response and the neutral mode

Every deterministic condition starts from its exact Figure 4 basal checkpoint. Uniform mean input is applied without noise and allowed to relax for 16,000 chemical time units or 1,600 Spring time units. A half-contrast edge must be unique, contrast at least 0.05, maximum native RHS no greater than 1e-6, and final 10% preparation change no greater than 1e-4 of the maximum state scale. These gates define a stationary polarized response; failure is recorded rather than replaced by another preparation.

The Gaussian pulse is centered at the interpolated rising or falling half-contrast crossing; its spatial standard deviation equals the declared noise covariance length ell (1.5 chemical, 0.15 Spring), and its duration is tau (50 chemical, 5 Spring). Peak a is 1e-4 of the unchanged native reference: 1e-5 Wave-pinning, 1e-4 Holmes, 4e-4 Spring, 6.4e-4 Goryachev. Each edge receives independent restarts at a/2, a and 2a. The other edge is tracked as well. A no-pulse continuation is subtracted at matching times. Spring edge locations use material fractions times mean-input circumference to remove uniform expansion.

A translated polarized profile is a neutral state. A brief pulse can therefore leave a persistent positional offset after the transient has relaxed: displacement does not have to return to zero. The plotted response is displacement divided by pulse amplitude. The inset is the time integral of that displacement through the explicitly finite 20-tau window, divided by the pulse dose a*tau. It is not an infinite-time integral. Terminal displacement divided by a*tau is the zero-frequency velocity gain.

A single spatial pulse does not identify susceptibility to every location on a ring. The additional spatial response shape is calculated from the unchanged native Jacobian at that same stationary mean-input state. Each edge kernel is normalized to unit gain under that exact pulse shape and multiplied by its measured terminal displacement per pulse dose. The unscaled native gain is separately validated against the measurement. No measured diffusion is used in this calculation. Both edges have the same zero-frequency translation susceptibility; transient shape/width modes need not be the same. The main prediction deliberately applies the requested independent-edge formula to those marginal edge diffusivities.

Linear-response definitions and exact imposed covariance

$$\delta\dot{\mathbf{y}} = J\,\delta\mathbf{y} + B\,\delta\mathbf{u}, \quad J\mathbf{q} = 0, \quad J^{\mathsf{T}}\mathbf{p} = 0, \quad \mathbf{p}^{\mathsf{T}}\mathbf{q} = 1. \tag{1}$$

$$\dot{s} = \sum_j g_j\,\delta u_j, \qquad g_j = (\mathbf{p}^{\mathsf{T}}B)_j. \tag{2}$$

$$K_u(j,k;t,t+v) = \mu^2\{\exp[\sigma^2 C_x(j-k)C_t(t,t+v)] - 1\}. \tag{3}$$

$$D_e = \sum_{j,k} g_j g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\,\mathrm{d}v. \tag{4}$$

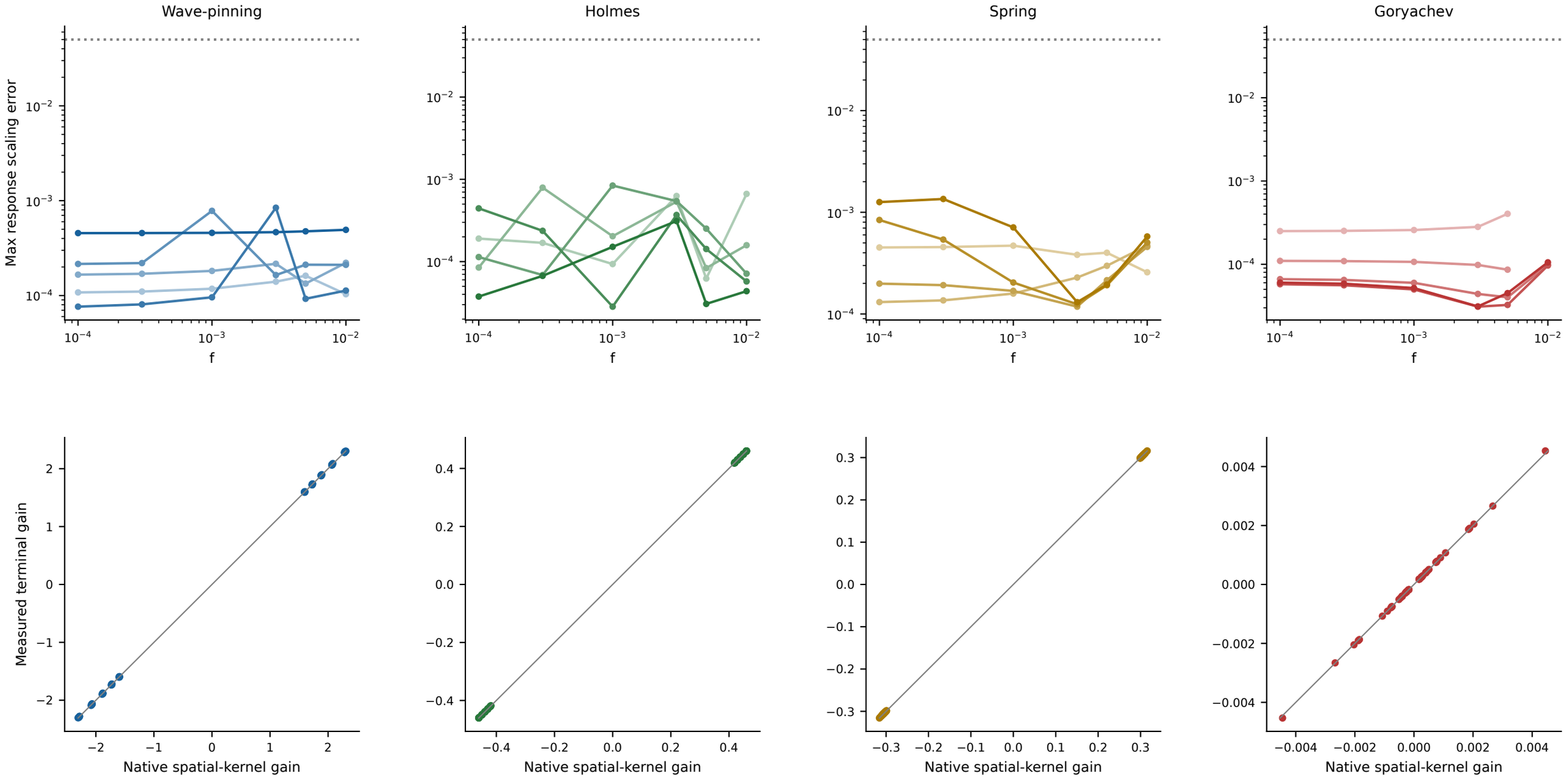
$$D_s^{\mathrm{ind}} = \tfrac{D_L + D_R}{4}, \qquad D_s = \tfrac{D_L + D_R + 2D_{LR}}{4}. \tag{5}$$

$$\mathcal{A}_{20\tau} = \tfrac{1}{a\tau}\int_0^{20\tau} [x_e^{\mathrm{pulse}}(t) - x_e^{\mathrm{control}}(t)]\,\mathrm{d}t. \tag{6}$$

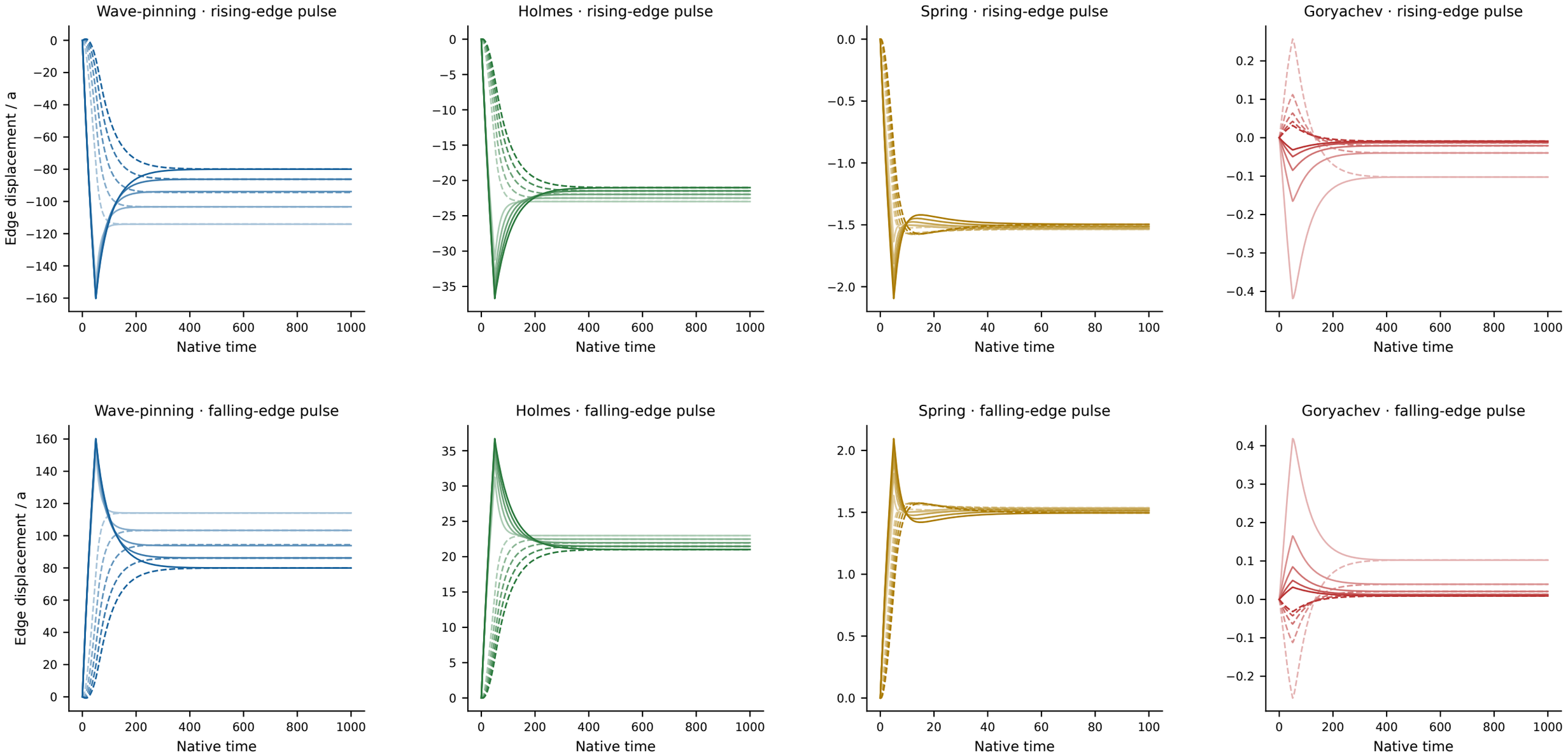
$$\widehat{g}_{e,j} = \tfrac{M_e}{Q_e} g_j, \quad M_e = \tfrac{\Delta x_e(20\tau)}{a\tau}, \quad Q_e = \sum_j g_j h_{e,j}. \tag{7}$$

q: derivative of the translated native profile; p: its adjoint, excluding conserved-pool modes.
Cx: discrete circular filter autocorrelation. Ct: analytically variance-normalized OU-knot interpolation.
The overbar averages the knot phase. Native sigma = 1; lognormal input covariance is not Gaussian OU covariance.

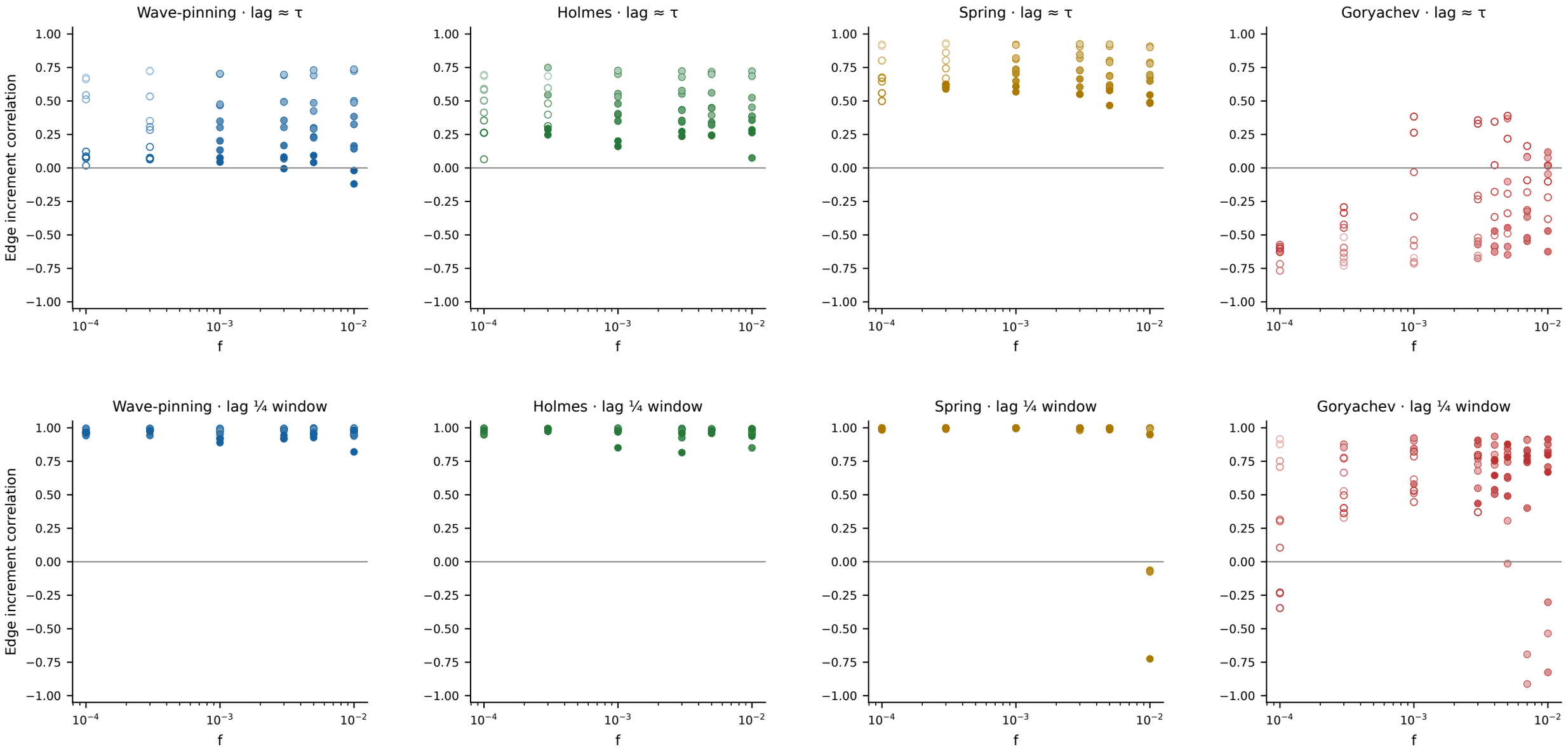
Amplitude checks and pulse validation; neither uses observed drift



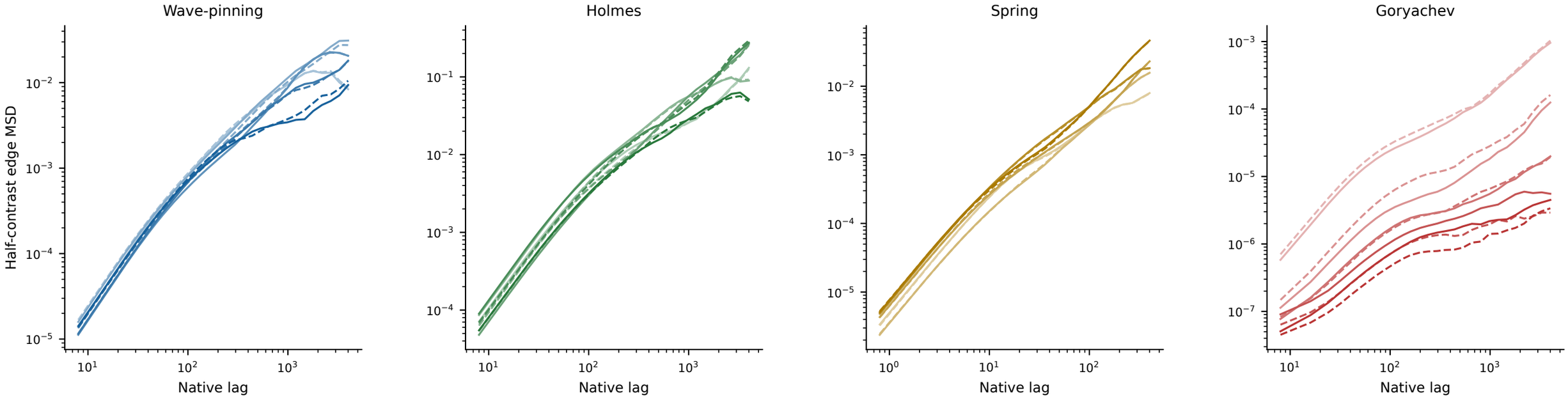
Both half-contrast edges move after a local pulse; f = 0.001



Solid: pulsed edge; dashed: opposite edge. Each shade is a Figure 4 size. A persistent common displacement is the neutral translation mode.

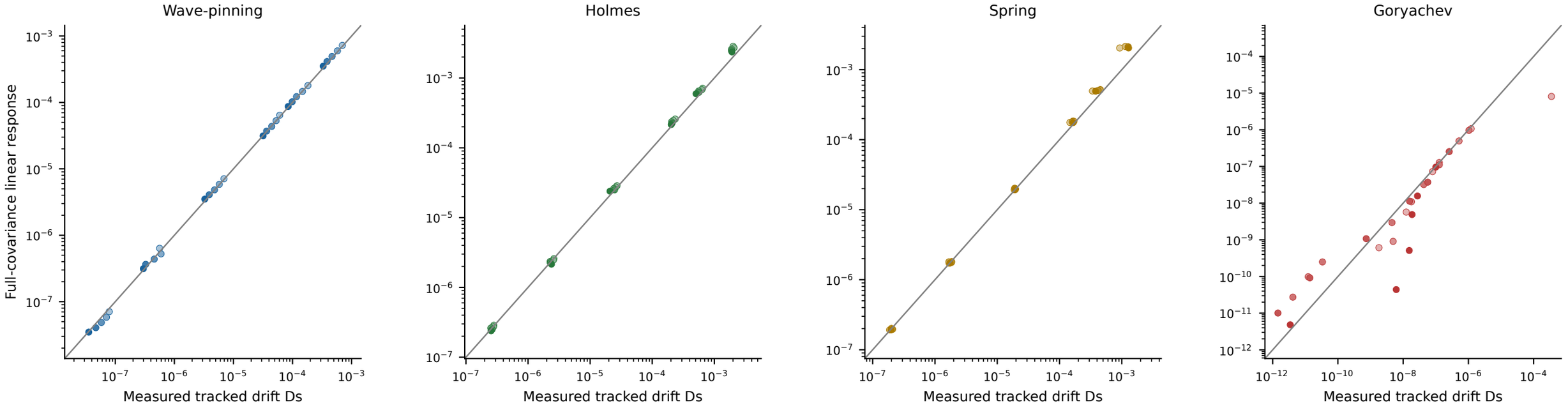

Independence check: Figure 4 exemplar seeds 0 and 1
Wave-pinning · lag ≈ τ
Holmes · lag ≈ τ
Spring · lag ≈ τ
Goryachev · lag ≈ τ
Wave-pinning · lag ¼ window
Holmes · lag ¼ window
Spring · lag ¼ window
Goryachev · lag ¼ window
Edge increment correlation
f
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
10^{-4}
10^{-3}
10^{-2}
Filled: both edge MSDs >10× conservative float16 crossing-error bound²; open: saved-profile precision may dominate.
Each point is a seed/size/input, only uninterrupted single-region pairs. These two exemplars do not give campaign confidence intervals.

Edge MSDs at f = 0.001; solid rising, dashed falling, seed 0



All seeds, fractions, pair counts, quantization bounds and covariance factors are provided in the plotted-data tables.

Full covariance sensitivity: retain the shared neutral translation



No parameter fit: the full correlated prediction is approximately twice the prescribed independent-edge prediction because the two measured-gain kernels nearly coincide.
This is a prediction of physical tracked drift, not an explanation of all heading decay, especially for Goryachev.